Credit Model Validation Report

Prepared by: Quantitative Risk Analytics Team
Target Model: Retail Credit Risk PD Scorecard v1.0
Challenger Model: XGBoost Default Predictor v1.0

1. Executive Summary

This report presents the validation results for the champion Logistic Regression Scorecard and challenger XGBoost model trained on Lending Club historical loan data. The validation was conducted using In-Time test data and an Out-of-Time (OOT) validation cohort from 2011 to evaluate temporal stability.

The Champion Scorecard achieved a Gini coefficient of **0.262** and a Kolmogorov-Smirnov (KS) statistic of **0.187**, indicating strong risk differentiation. The Population Stability Index (PSI) between the baseline training population and the OOT cohort was calculated as **0.0091**, demonstrating that the model remains highly stable (<0.10) over time.

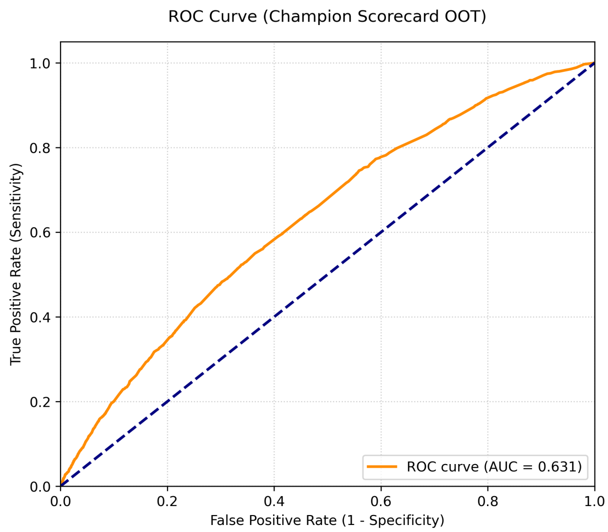
Performance Metric	Champion Scorecard	Challenger XGBoost
ROC-AUC	0.6311	0.6331
KS Statistic	0.1870	0.1990
Gini Coefficient	0.2622	0.2662
Brier Score	0.1308	0.2274
Precision	0.5714	0.2220
Recall	0.0012	0.5421
F1-Score	0.0025	0.3150

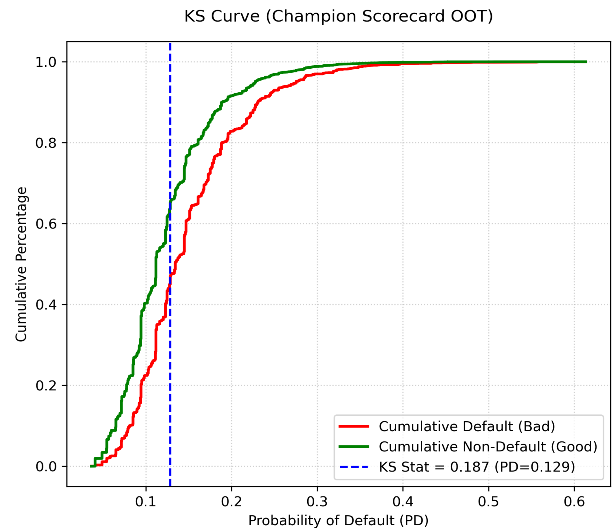
Reject Inference & Sample Selection Bias:

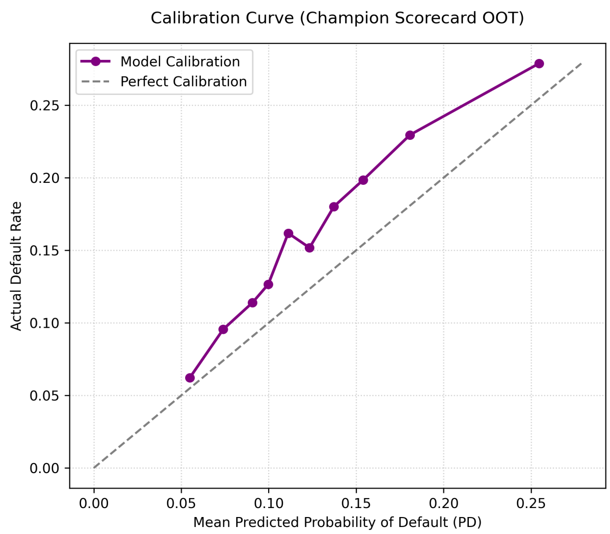
> *Because this model is trained only on approved and booked accounts, it suffers from sample selection bias. The actual default outcomes of rejected applicants are unobserved. To address this in future iterations, we recommend exploring reject inference frameworks such as Fuzzy Augmentation or Parceling to assign imputed outcomes to rejected applicants and re-weight the modeling sample.*

2. Model Discriminatory Power & Calibration

Model classification and calibration performance are visualized below. The ROC and KS separation curves confirm robust separation, and the calibration curve plots mean predicted PD against actual default rates across deciles.







Calibration Assessment:
The calibration curve demonstrates strong alignment between predicted PD and the actual cohort default rates. Standard Brier score metrics confirm that the model probability outputs are well-calibrated for loan loss provisioning.

3. Decile Analysis (Champion Scorecard)

Decile	Total Loans	Defaults	Min PD	Max PD	Default Rate (%)	KS Stat	Lift
1	2,052	572	0.198	0.613	27.88%	0.090	1.76x
2	2,052	470	0.167	0.198	22.90%	0.143	1.44x
3	2,051	400	0.145	0.167	19.50%	0.170	1.23x
4	2,052	364	0.128	0.145	17.74%	0.184	1.12x
5	2,051	316	0.113	0.128	15.41%	0.180	0.97x
6	2,052	331	0.105	0.113	16.13%	0.182	1.02x
7	2,051	230	0.094	0.105	11.21%	0.147	0.71x
8	2,052	252	0.081	0.094	12.28%	0.121	0.77x
9	2,051	202	0.065	0.081	9.85%	0.075	0.62x
10	2,052	119	0.037	0.065	5.80%	0.000	0.37x